

EXECUTIVE SUMMARY

Comparative Study of Speech-to-Text Models for Noisy Real-World Audio

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Objective

This study compares three automatic speech recognition (ASR) configurations for a voice-based customer-support workflow: OpenAI Whisper Base, Faster-Whisper Base, and Wav2Vec2 Base-960h. The evaluation considers recognition error, CPU inference latency, real-time processing capability, memory consumption, deployment practicality, and robustness to controlled acoustic noise.

Dataset and Method

A fixed subset of 100 Hindi utterances from Mozilla Common Voice was evaluated under two conditions: original clean speech and the same utterances with additive white Gaussian noise at 10 dB signal-to-noise ratio. Audio was processed at 16 kHz mono, with batch size one. One untimed warm-up inference preceded latency measurement, while model download and initialization time were excluded. The benchmark recorded Word Error Rate (WER), inference time, real-time factor (RTF), and CPU memory usage.

Key Benchmark Results

Model	Clean WER	Noisy WER	Clean Latency	CPU Memory
Whisper Base	1.1018	1.1251	4.416 s	840.1 MB
Faster-Whisper Base	1.0857	1.1057	2.336 s	619.7 MB
Wav2Vec2 Base-960h	1.0748	1.1847	0.376 s	962.9 MB

Findings and Interpretation

Wav2Vec2 produced the lowest measured clean WER and the lowest CPU latency (0.376 s; RTF 0.083), but its WER increased from 1.0748 to 1.1847 under noise, a degradation of 0.1099. The selected Base-960h checkpoint is an English-only ASR model, while the benchmark references are Hindi in Devanagari; therefore, its measured WER reflects a language/checkpoint mismatch as well as acoustic effects and should not be interpreted as native Hindi ASR accuracy.

Faster-Whisper showed the most balanced measured profile in this CPU experiment. Its clean and noisy WER values were 1.0857 and 1.1057, giving the smallest measured noise-induced increase (0.0200). It also reduced clean inference latency to 2.336 s and CPU memory to 619.7 MB compared with standard PyTorch Whisper (4.416 s and 840.1 MB). Whisper retained broader multilingual capability but required more CPU time and memory in this configuration.

Production Implication

For the tested Hindi customer-support scenario, Faster-Whisper is the most directly aligned configuration because it combines the multilingual Whisper model family with a more efficient inference runtime and the smallest measured noise-induced WER increase. This is a scenario-specific engineering recommendation, not a universal ranking. Before production deployment, the configuration should be validated on representative call-center audio, a larger language-appropriate test set, and the intended serving hardware.

Next-Step Strategy

A production adaptation should use domain-specific call-center speech with speaker-level train/validation/test separation, noise and telephony-channel augmentation, and coverage of customer-support terminology. Practical optimization can include INT8 quantization, voice activity detection, efficient serving, and concurrency-aware batching. The proposed production flow is: audio ingestion -> preprocessing/noise filtering -> voice activity detection -> ASR inference -> transcript post-processing -> downstream AI and monitoring.